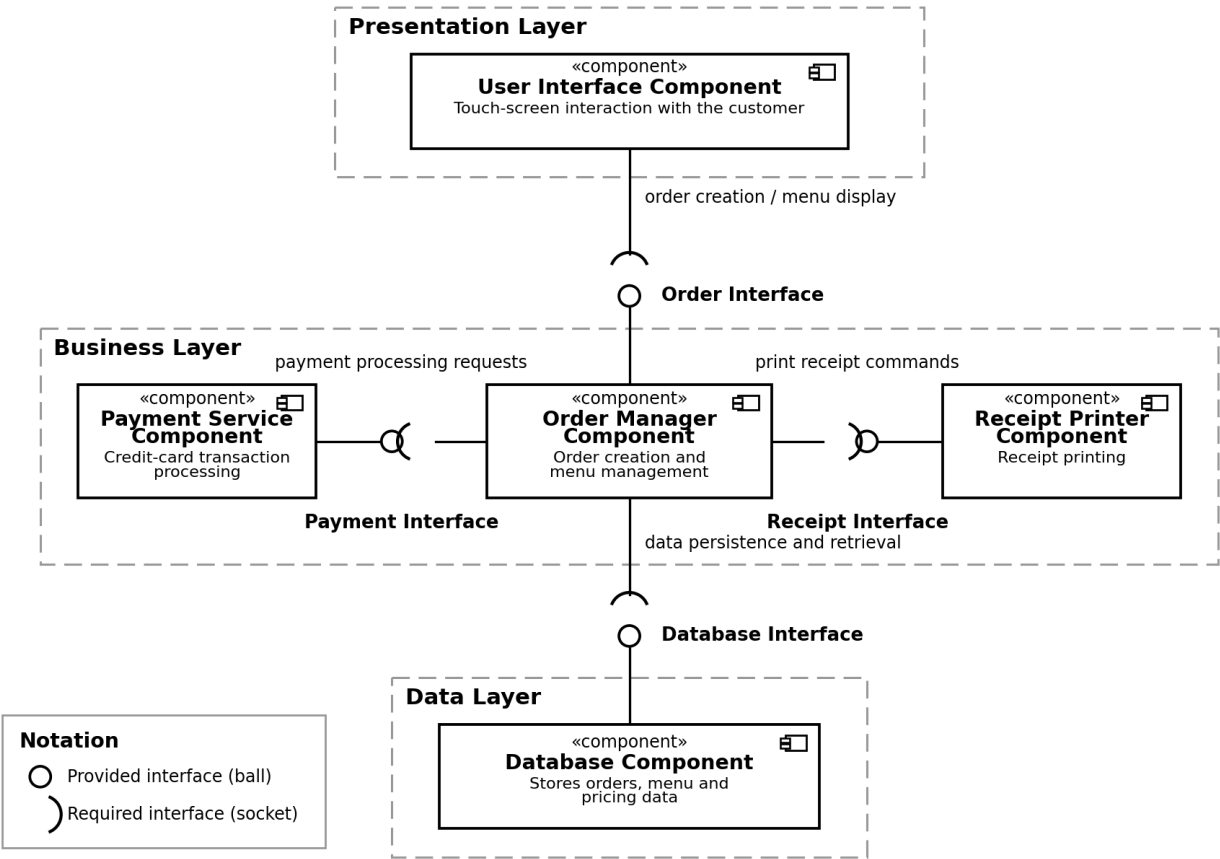


# Self-Service Coffee Kiosk System

## Component Diagram

Self-Service Coffee Kiosk System - Layered Architecture



# Architecture Selection and Justification

## Lab 3 — Self-Service Coffee Kiosk System

### Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

#### Reason 1: Clear separation of responsibilities

The kiosk has different responsibilities that can be separated into presentation, business and data layers. The User Interface Component handles the touch-screen interaction, the Order Manager, Payment Service and Receipt Printer handle the main ordering functions, and the Database Component stores menu, pricing and order data. This makes the system easier to understand and keeps each component focused on its role.

#### Reason 2: Suitable for the kiosk requirements

The kiosk needs a simple ordering flow from selecting a coffee and size to making a credit-card payment and printing a receipt. Layered Architecture provides a clear flow between these functions through the Order Manager, while also keeping database operations separate from the user interface. This fits the focused set of functions required by the kiosk.

#### Security Advantage

The payment functionality is kept in a separate Payment Service Component instead of being handled directly by the user interface. This separates the customer-facing part of the system from the component responsible for credit-card transaction processing, which helps limit where payment-related processing is performed.

#### Performance Benefit

The architecture keeps the ordering flow structured and direct: the User Interface communicates with the Order Manager, which then handles payment, receipt printing and data access. This avoids unnecessary architectural complexity and supports the requirement that an order should be completed in under 60 seconds.